

## Markscheme

Mathematics: analysis and approaches  
Practice question bank

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**86. (a)**

Consider the statement: "for every positive integer  $n$ , the expression  $n^2 - n + 41$  is a prime number."

Find the smallest positive integer  $n$  for which this statement is false.

test small values of  $n$ :  $n = 1, 2, 3, \dots$  all give prime results, so no small counter-example exists *(M1)*

consider  $n = 41$ :  $41^2 - 41 + 41 = 41^2$  *(M1)* **A1**

since  $41^2 = 41 \times 41$  has a factor other than 1 and itself, it is not prime **R1**

the smallest counter-example is  $n = 41$  **A1**

**[5 marks]**